



LDPolypVideo Benchmark: A Large-Scale Colonoscopy Video Dataset of Diverse Polyps

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Abstract. Computer-Aided Diagnosis (CAD) systems for polyp detection provide essential support for colorectal cancer screening and prevention. Recently, deep learning technology has made breakthrough progress in medical image computation and computer-aided diagnosis. However, the deficiency of training data seriously impedes the development of polyp detection techniques. Existing fully-annotated databases, including CVC-ClinicDB, ETIS-Larib, CVC-Colon dataset, Kvasir-Seg, and CVC-ClinicVideoDB, are very limited in polyp size and shape diversity, which is far from the significant complexity in the actual clinical situation. In this paper, we propose LDPolypVideo, a large-scale colonoscopy video database that contains a variety of polyps and more complex bowel environments. Our database contains 160 colonoscopy videos and 40,266 frames in total with polyp annotations, which are four times the size of the largest existing colonoscopy video database CVC-ClinicVideoDB. In order to improve the efficiency of polyp annotation, we design an intelligent annotation tool based on object tracking. Extensive experiments have been conducted to evaluate state-of-the-art object detection approaches on our LDPolypVideo dataset. The average drops of Recall and Precision of four SOTA approaches on this dataset are 26% and 15%, respectively. The great performance drop demonstrates the significant challenges but also the great value of our large-scale and diverse polyp video dataset to facilitate the research on polyp detection. Our dataset is available at <https://github.com/dashishi/LDPolypVideo-Benchmark>.

Keywords: Colonoscopy video · Large-scale dataset · Polyp detection · Tracking-assisted annotation

1 Introduction

Colorectal cancer (CRC) is one of the leading causes of death worldwide [17]. Polyps are considered as precursors to colorectal cancer. Early detection of

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polyps can efficiently reduce the cancer rate. Video colonoscopy is now commonly used for polyp detection. However, the detection rate of polyps highly depends on the doctor’s skill and experience on colonoscopy screening. Polyps are easily missed due to complex intestinal environments, specular reflections, and limited field of view.

In order to improve the detection rate of polyps, the computer-aided diagnosis has attracted great attention from researchers. Many approaches have been proposed to apply computer vision algorithms to the automatic detection of polyps in colonoscopy images or videos. Earlier studies explore hand-crafted features including shape, color, and texture for polyp detection and diagnosis [1, 20, 23, 24]. Recently, convolutional neural networks (CNN) have gained much more attention in medical image computation. Utilizing common object detection networks with colonoscopy imagery, many polyp detection approaches have been developed [5, 19, 21, 22]. However, learning-based approaches significantly rely on training data. Acquiring large-scale colonoscopy images with accurate polyp annotation is arduous due to the large diversity of polyps and the requirement of expertise. A few small-scale datasets that contain high-quality colonoscopy images have been built for polyp detection in single colonoscopy images [2, 4, 18]. Hyper-Kvasir [6] is the largest dataset for gastrointestinal disease detection. It contains 73 videos that are classified as polyp but no localization annotations of the polyps. Thus it does not support supervised methods for polyp localization. Among it, a subset of 1000 noncontinuous polyp images, called as Kvasir-Seg, contain polyp masks and can be used for polyp localization.

Real clinical colonoscopy videos, however, have more diverse appearances and lower image quality with motion blurs, which pose significant challenges for these CNN-based models on accurate and continuous polyp detection. In order to improve the detection rate of polyps in colonoscopy videos, some studies exploit temporal correlations to guide continuous polyp detection [13, 26, 27]. These approaches have achieved promising results on existing colonoscopy video datasets that are not large and diverse enough for both training and comprehensive evaluation. We make an extensive survey on existing colonoscopy datasets for polyp detection and list the parameters in Table 1. The widely used video dataset, CVC-ClinicVideoDB, only contains 18 videos with annotations and 18

Table 1. Summary of public annotated colonoscopy datasets. Our LDPolypVideo dataset is the largest one up to now.

Dataset	Format	Resolution	N_{images}	N_{videos}	N_{polyps}
CVC-ClinicDB [2]	Image	384×288	612	29	29
CVC-ColonDB [4]	Image	574×500	380	15	15
ETIS-Larib [18]	Image	1225×966	194	34	44
Kvasir-Seg [6]	Image	Various	1000	N/A	N/A
CVC-ClinicVideoDB [1, 3]	Video	384×288	11945	18	18
LDPolypVideo	Video	560×480	40266	160	200

videos without polyp annotations. Besides, there is at most one polyp in each frame of the CVC-ClinicVideoDB dataset. In contrast, it is a common case that multiple polyps occur at the same time in clinical examination.

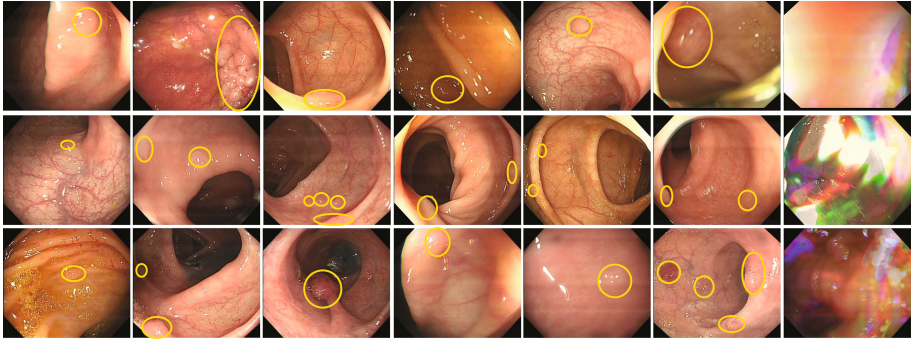


Fig. 1. Examples of images and polyp annotations in our LDPolypVideo dataset. The last column shows a few challenging frames with motion blur.

In order to increase the size and diversity of colonoscopy data for training and evaluation of polyp detection approaches, we present a large-scale and diverse colonoscopy video dataset named LDPolypVideo. It consists of 160 videos with 40,266 frames, nearly four times the size of the largest existing fully-annotated dataset. There are 33,884 frames that contain at least one polyp and in total 200 labeled polyps, which are more than 11 times the polyps in CVC-ClinicVideoDB. The polyps present more diverse morphologies. Figure 1 shows a set of example images and the annotated masks for polyps. Besides, we also provide 103 videos, including 861,400 frames without full annotations. Each video has a label indicating whether it contains polyps. These videos enrich the data diversity and will support unsupervised and semi-supervised methods. Based on our LDPolypVideo dataset, we evaluate a number of state-of-the-art approaches for polyp detection to analyze their strengths and weaknesses, demonstrating the challenges of colonoscopy polyp detection in clinical examination.

In summary, our contributions mainly are two-folds:

1. We propose a large-scale colonoscopy video dataset, LDPolypVideo, about four times the size of the existing largest database. We provide a better benchmark for polyp detection.
2. In order to improve the efficiency of data labeling, we develop an intelligent labeling tool based on a video tracking algorithm, which greatly reduces the labeling effort and can be used for other datasets.
3. Based on our LDPolypVideo dataset, we conduct extensive experiments to analyze the challenges of polyp detection in practical clinical data.

2 Our LDPolypVideo Dataset

Our goal to establish a colonoscopy polyp dataset consists of two key points, **large scale** and **diversity**. A large-scale dataset can meet the demand of deep learning-based studies and avoid over-fitting. Meanwhile, great diversity will improve the practical significance of trained models in the clinic.

2.1 Data Acquisition

In order to get a large-scale dataset, we collected clinical colonoscopy videos from routine clinical examinations at a state hospital. All metadata about the patients were removed. An experienced clinician selects video clips that contain polyps. After cutting off the black borders, we resize the video frames to 560×480 . We sampled 160 videos with 40,266 frames while ensuring that there is at least one polyp in each video. There are in total 33,884 frames that contain at least one polyp. Figure 2 shows the statistical information about our dataset, including the number of frames (blue bars), the number of positive frames where there is at least a polyp (orange bars), and the total number of polyps in each video.

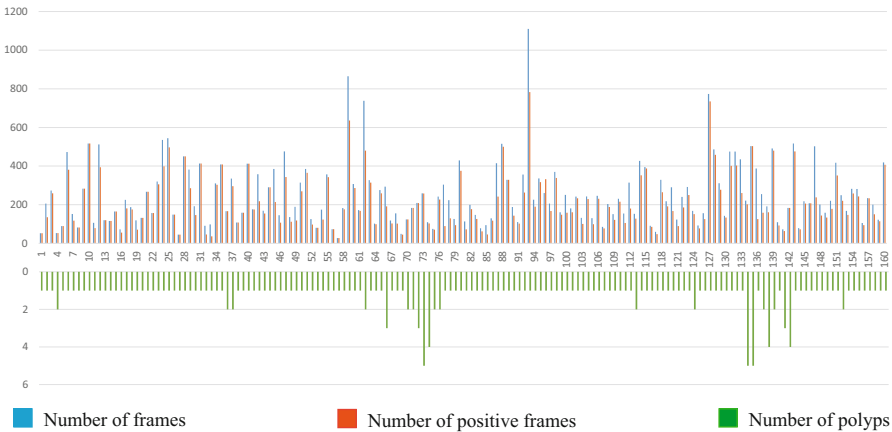


Fig. 2. Summary of our LDPolypVideo dataset.

We increase the diversity of colonoscopy images in two aspects. First, motion blur caused by camera movement, colon folds, and intestinal peristalsis frequently occur in clinical colonoscopy. However, previous colonoscopy image datasets only contain high-quality images and distinct polyps. As Fig. 3 shows, motion blurs and specular reflections bring great disturbance for polyp detection. In the second row, the polyp appears in continuous frames of a video but is difficult to detect at the beginning due to colon folds. In this case, temporal context plays a critical role for accurate detection. Second, there might be multiple

polyps in a single frame. In our LDPolypVideo dataset, 22 videos contain multiple polyps, as shown in Fig. 1 and Fig. 2. Particularly, three videos in our dataset contain five polyps, which are very challenging to completely detect, even for an experienced clinician. In this situation, a robust computer-aided polyp detection tool is valuable to improve the detection rate in polyp screening.

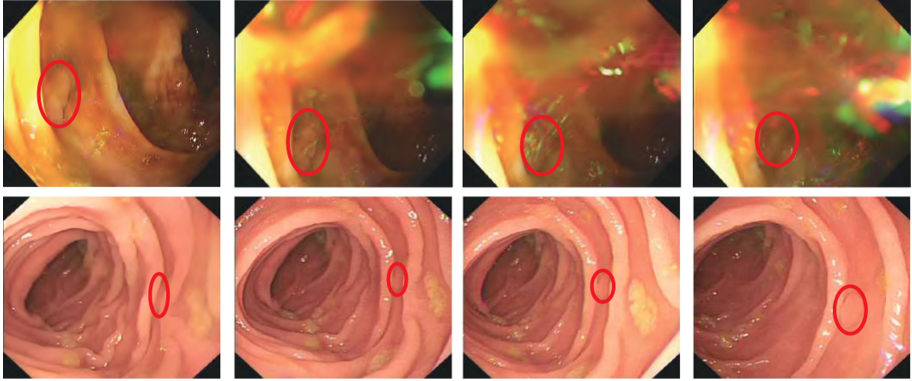


Fig. 3. Video frames with motion blur and reflections (top) and a polyp example that is difficult to detect due to colon folds.

2.2 Tracking-Assisted Annotation Tool

The most labor-intensive but important work in establishing a database is polyp annotation in all frames. Instead of asking experts to manually label the polyps frame by frame, we develop an intelligent annotation tool by combining manual labeling with object tracking algorithms. Considering efficiency and accuracy, we choose the Kernelized correlation filtering method [11] as the tracker. It trains a correlation filter from two adjacent frames and calculates the correlation of a frame with the previous labeled frame to predict the presence probability of the labeled object. We select the point with the highest probability as the candidate position in the new frame. Utilizing this tracker, for each video, we ask an expert to label a polyp when it first appears and the tracker automatically predicts its position in the next frame. The expert only needs to check whether the tracking result is acceptable. If the tracking result is not satisfied, manual annotation is performed on this new frame and tracking continues. Figure 4 illustrates our annotation process. After a round of labeling for the entire video, we ask another expert to verify the annotations and make modifications if necessary. This tracking-assisted annotation pipeline is very effective and efficient. We tested our tool on two videos in HyperKvasir. For the 894 frames in these two videos, the polyps in 681 frames can be automatically annotated and only 80 frames require manual re-annotation, which shows the usefulness of our tool on other datasets.

3 Experiments on LDPolypVideo Dataset

The large size and diversity of our LDPolypVideo dataset pose challenges for polyp detection. We conduct a series of experiments to comprehensively evaluate state-of-the-art (SOTA) detection approaches and demonstrate the challenges of clinical colonoscopy polyp screening.



Fig. 4. The process of tracking-assisted polyp annotation in colonoscopy videos.

Table 2. Performance of recent polyp detection approaches on existing datasets.

Dataset	Method	Baseline	Recall	Precision	F1	F2
ETIS-Larib	Shin et al. [16]	Faster RCNN	0.80	0.73	0.78	0.78
	Wittenberg et al. [25]	Mask RCNN	0.80	0.87	0.82	0.82
	Brandao et al. [7]	FCN	0.83	0.74	0.81	0.81
	Sornapudi et al. [19]	FPN+RCNN	0.74	0.77	0.74	0.74
CVC-ColonDB	Sornapudi et al. [19]	FPN+RCNN	0.92	0.90	0.97	0.91
	Brandao et al. [7]	FCN	0.90	0.80	0.85	0.88
CVC-ClinicDB	Wang et al. [22]		0.88	0.93	0.91	0.89
	Wittenberg et al. [25]	Mask RCNN	0.86	0.80	0.82	0.85
ASU-Mayo	Yu et al. [26]	3D-FCN	0.71	0.88	0.78	0.73
	Zhang et al. [27]	Yolo	0.72	0.89	0.9	0.74
	Shin et al. [16]	Faster RCNN	0.84	0.83	0.83	0.84
CVC-ClinicVideo	Shin et al. [16]	Faster RCNN	0.84	0.89	0.87	0.85
	Qadir et al. [13]	Faster RCNN	0.81	0.87	0.84	0.83

We first summarize recent approaches for colonoscopy polyp detection on existing datasets in Table 2. Generally, the higher performance is achieved for polyp detection on the colonoscopy image datasets CVC-ColonDB [4] and CVC-ClinicDB [2] compared to the results on video polyp detection. On the video dataset CVC-ClinicVideo [1, 3], Both Shin et al. [16] and Qadir et al. [13] use Faster R-CNN [15] as the backbone network. The former adopts post learning to fine-tune the detector with detected positives in the test video and the latter exploits temporal dependency between frames to reduce the false positives.

Following these methods, we evaluate four commonly used object detection networks, including Faster R-CNN [15], Yolo3 [14], RetinaNet [12], and Center-Net [9], on our LDPolypVideo dataset. We choose ResNet-50 [10] as the backbone

for Faster R-CNN, RetinaNet, and CenterNet, while Yolo3 adopts DarkNet [14] as its backbone. We do not implement any post-training on specific videos for a fair comparison. We train all detection networks on CVC-ClinicDB, which consists of 612 colonoscopy images with single polyps. To prevent overfitting, data augmentation including image rotation, flipping, scaling, shearing, and motion blur is performed. All the models are pre-trained on ImageNet [8] and then fine-tuned on the augmented dataset. We test these four trained detection models on both CVC-ClinicVideo and our LDPolypVideo dataset. The results are shown in Table 3. Overall, RetinaNet performs the best on the two datasets. Compared to CVC-ClinicVideo which only contains 18 videos, the performance of these approaches significantly degrades on our LDPolypVideo dataset. The average drops of Recall and Precision are 26% and 15%, respectively.

Table 3. Performance comparison on CVC-ClinicVideo and LDPolypVideo datasets.

Methods	CVC-ClinicVideo				LDPolypVideo			
	Recall	Precision	F1	F2	Recall	Precision	F1	F2
Faster RCNN [15]	0.61	0.83	0.70	0.65	0.30	0.68	0.42	0.34
Yolo3 [14]	0.45	0.82	0.58	0.49	0.29	0.67	0.41	0.33
RetinaNet [12]	0.64	0.85	0.73	0.67	0.47	0.65	0.55	0.50
CenterNet [9]	0.57	0.98	0.72	0.62	0.17	0.88	0.28	0.20

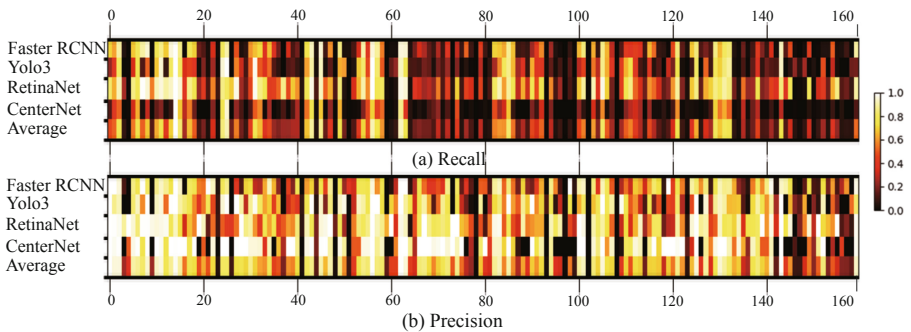


Fig. 5. Recall and precision of four detection networks on 160 videos in our dataset. Darker colors represent poorer performances.

In order to analyze the challenges of our LDPolypVideo dataset for SOTA models trained on colonoscopy image datasets, we analyze their performance on each individual video and visualize the precision and recall in Fig. 5. It shows that the performance varies in a wide range for the 160 videos. We select several typical challenging cases and analyze their common characteristics. Figure 6 shows nine representative cases. For the first two cases (a)(b), the small polyps are very difficult to detect. None of the four detectors succeed in detecting them

at any frame. For the cases (c) to (f), the four detectors fail to detect the polyps in successive frames because of specular reflections, small size, motion blur, colon folds, etc. The polyp in (c) is not detected when it is partially visible with indistinguishable appearances with the background. In case (d), a colorectal ileocecal valve is detected by Yolo3 by mistake. In cases (e) and (g), there are multiple polyps in the same video. They are very challenging for the detectors to completely localize them, especially for small polyps. We also show typical false positives in Fig. 6 (h)(i). All detectors are easily deceived by the colorectal ileocecal valves and colon folds which look similar to polyps.

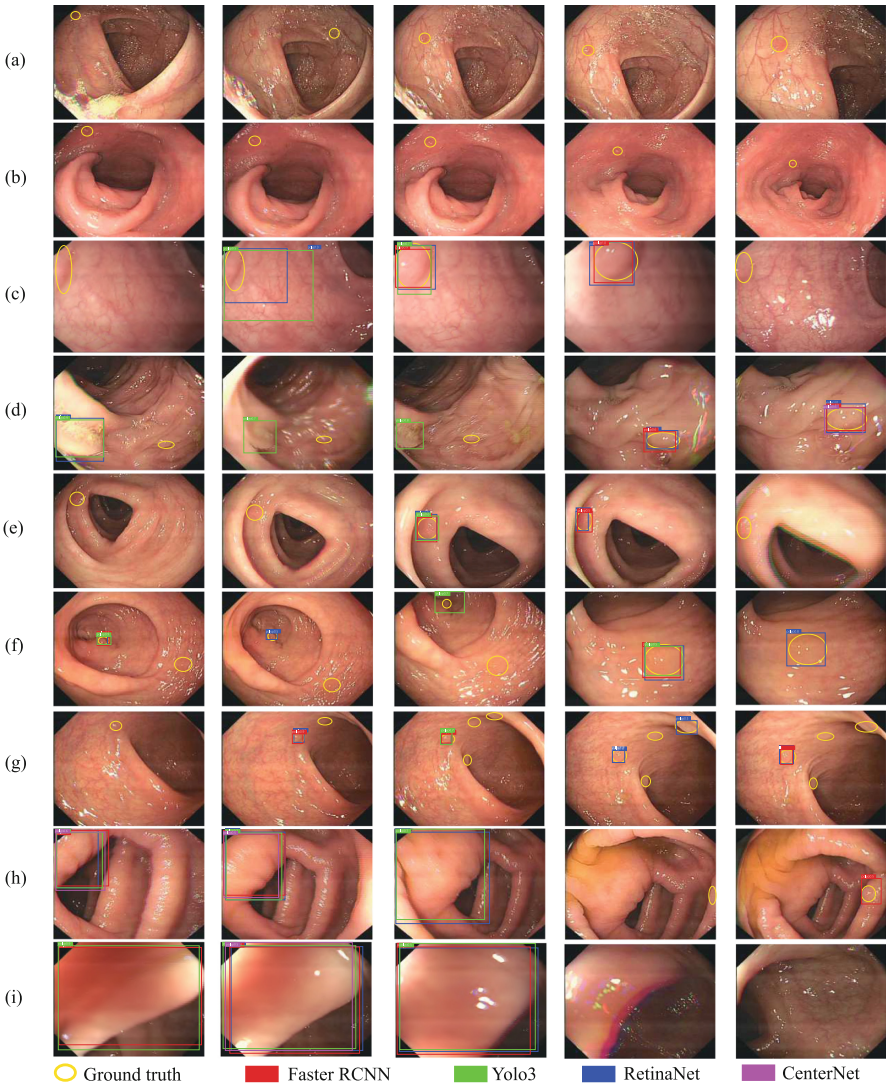


Fig. 6. Representative cases in our dataset and the detection results of SOTA methods.

4 Conclusion

In this paper, we present a large-scale colonoscopy video dataset, named LDPolypVideo, with polyp annotations in all frames. This new dataset contains 160 colonoscopy videos that are decomposed into 40,266 frames with 200 polyps. Among them, there are 33,884 frames that contains at least one polyp. We developed an efficient labeling tool based on object tracking to facilitate the labeling process. The large dataset size, great diversity in polyp morphology, multiple polyps in single frames, motion blur, and specular reflections in the colonoscopy videos pose significant challenges for polyp detection. We conducted extensive experiments to evaluate a number of state-of-the-art detection approaches and comprehensively analyze the characteristics of our dataset. We believe that our LDPolypVideo dataset provides a new benchmark and will facilitate researches on computer-aided polyp detection and diagnosis.

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